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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 7	
TITLE: Inheritance and Interfaces	

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

1.Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

Programme

Assignment6.java | Assignment7.java

src > Assignment7.java > ...

```java
class Shape {
    double area() {
        return 0;
    }
}

class Circle extends Shape {
    double radius;

    Circle(double radius) {
        this.radius = radius;
    }

    double area() {
        return Math.PI * radius * radius;
    }
}

class Rectangle extends Shape {
    double length, breadth;

    Rectangle(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }

    double area() {
        return length * breadth;
    }
}

public class Assignment7 {
    Run | Debug
    public static void main(String[] args) {

        Circle c = new Circle(radius: 5);
        Rectangle r = new Rectangle(length: 10, breadth: 5);

        System.out.println("Area of Circle: " + c.area());
        System.out.println("Area of Rectangle: " + r.area());
    }
}
```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                    powershell
PS C:\Users\RIKISHA\PROGRAMMING\Java> cd src
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac Assignment7.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java Assignment7
Area of Circle: 78.53981633974483
Area of Rectangle: 50.0
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Programme

Assignment6.java | Assignment8.java | EcommerceDemo.java

src > EcommerceDemo.java > EcommerceDemo

```java
interface Product {
    void displayProduct();
}

class ElectronicProduct implements Product {
    String name = "Laptop";
    double price = 55000;

    public void displayProduct() {
        System.out.println("Electronic Product: " + name);
        System.out.println("Price: Rs. " + price);
    }
}

class ClothingProduct implements Product {
    String name = "T-Shirt";
    double price = 1200;

    public void displayProduct() {
        System.out.println("Clothing Product: " + name);
        System.out.println("Price: Rs. " + price);
    }
}

class GroceryProduct implements Product {
    String name = "Rice";
    double price = 800;

    public void displayProduct() {
        System.out.println("Grocery Product: " + name);
        System.out.println("Price: Rs. " + price);
    }
}

public class EcommerceDemo {
    Run | Debug
    public static void main(String[] args) {

        Product p1 = new ElectronicProduct();
        Product p2 = new ClothingProduct();
        Product p3 = new GroceryProduct();

        p1.displayProduct();
        p2.displayProduct();
        p3.displayProduct();
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac EcommerceDemo.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java EcommerceDemo
Electronic Product: Laptop
Price: Rs. 55000.0
Clothing Product: T-Shirt
Price: Rs. 1200.0
Grocery Product: Rice
Price: Rs. 800.0
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

Conclusion

This assignment helped us understand interfaces and inheritance in Java. We learned how a class can implement and inherit features from another interface. The program demonstrates code reusability, abstraction, and polymorphism in Java.

GitHub Screenshot